

Question number	Answer	Mark
8(a)(i)	D nm The only correct answer is D. <i>A is incorrect because cm would be 1×10^0</i> <i>B is incorrect because mm would be 1×10^{-1}</i> <i>C is incorrect because μm would be 1×10^{-4}</i>	(1)

Question number	Answer	Mark
8(a)(ii)	B Q and T The only correct answer is B. <i>A is incorrect because R is a fatty acid and only contains C, H and O</i> <i>C is incorrect because R is a fatty acid and only contains C, H and O, S is cholesterol and only contains C, H and O</i> <i>D is incorrect because S is cholesterol and only contains C, H and O</i>	(1)

Question number	Answer	Mark
8(a)(iii)	D S : T The only correct answer is D. <i>A is incorrect because</i> <i>B is incorrect because</i> <i>C is incorrect because</i>	(1)

Question number	Answer	Additional guidance	Mark
8(b)	An explanation that includes the following points: - because the sequence of amino acids determine the {tertiary / quaternary} structure of the protein (1) - by determining the {position / type} of bonds that form between the <u>R groups</u> (1) {hydrophobic / non-polar} {(R) groups / amino acids on (the outside of) the part of the protein that is embedded in the fatty acid tails (1) {hydrophilic / polar} {(R) groups / amino acids} (on the outside of) the part of the protein that is {amongst the phosphate heads / facing the cytoplasm / facing the aqueous environment} (1)	ACCEPT pieced together ACCEPT a named bond between R groups	(4)

Question number	Answer
*8(c)	Indicative content: Molecule E • enters by osmosis • because it is water • the number of the molecules F, G and H lower the water potential • so E moves down the water potential gradient Molecule F • enters by facilitated diffusion • down its concentration gradient • because it is polar and cannot pass through the fatty acid tails • therefore needs a protein to provide a polar channel for it to diffuse through Molecule G • enters by diffusion • because it is at a higher concentration outside the cell • and it is non-polar so can pass through the fatty acid tails Molecule H • enters by active transport • because it is at a higher concentration inside the cell • and therefore needs ATP and a protein to pump it across the membrane Level 1 : 1 mark = 1 relevant comment 2 marks = 3 relevant comments Level 2 : 3 marks = 4 relevant comments for at least two molecules 4 marks = 5 relevant comments for at least two molecules Level 3 : 5 marks = 6 relevant comments for at least three molecules 6 marks = 7 relevant comments from all four molecules (6)